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Calculators

	Section	Count
1	Products	8
2	Industries	18
3	Manufacturing Types	5
4	Problem Pages	8
5	Personas	9
6	**Total**	396 calculators

Products (8 modules)

	Code	Full Form
1	**PM-PMS**	Production Monitoring System
2	**PM-CMS**	Condition Monitoring System
3	**PM-CT**	Control Tower
4	**PM-QMS**	Quality Management System
5	**PM-TRC**	Traceability System
6	**PM-EMS**	Energy Monitoring System
7	**PM-AT**	Assembly Tracking
8	**PM-PF**	Platform (User, Role, Assets, Notifications etc.)

Industries (18)

	Code	Full Form
1	**I-Auto**	Automotive & Auto Components
2	**I-FMCG**	FMCG & Food Processing

3	**I-Pharma**	Pharma & Life Sciences
4	**I-Text**	Textile & Garments
5	**I-Elec**	Electronics Manufacturing
6	**I-Plas**	Plastics & Injection Molding
7	**I-Heavy**	Heavy Engineering / Forge / Foundry
8	**I-OEM**	Machine Builders (OEM)
9	**I-Fab**	Fabrication & Metalworking
10	**I-Aero**	Aerospace & Defense
11	**I-Med**	Medical Devices
12	**I-CD**	Consumer Durables & Home Appliances
13	**I-Rub**	Rubber & Tyre Manufacturing
14	**I-Steel**	Steel & Metal Mills
15	**I-Pack**	Paper & Packaging
16	**I-IEq**	Industrial Equipment & Capital Goods
17	**I-Chem**	Chemical & Paint
18	**I-EV**	Battery & EV Components

Manufacturing Types (5)

	Code	Full Form
1	**M-Disc**	Discrete Manufacturing
2	**M-Proc**	Process Manufacturing
3	**M-Batch**	Batch Manufacturing
4	**M-Asm**	Assembly-Line Manufacturing
5	**M-Util**	Utility / Energy-Intensive Operations

Problem Statement Pages (8)

	Code	Full Form
1	**PR-DT**	High Downtime
2	**PR-SR**	High Scrap & Rejection
3	**PR-LV**	Low Production Visibility
4	**PR-QT**	Quality Issues
5	**PR-TR**	No Traceability
6	**PR-EN**	High Energy Cost
7	**PR-AE**	Assembly Errors
8	**PR-SC**	Hard to Scale Digitalization

Personas (9)

	Code	Full Form
1	**PE-CXO**	CEO / CFO / COO
2	**PE-PH**	Plant Head
3	**PE-MT**	Maintenance Head
4	**PE-QT**	Quality Head
5	**PE-PR**	Production Supervisor
6	**PE-ESG**	Sustainability / ESG Head
7	**PE-IT**	CIO / IT Lead
8	**PE-OP**	Operator
9	**PE-NTI**	New to IIoT / Beginner Persona

	Name	Description	Which Page	Inputs	Outputs	Formula	Relevant Calculators	What & Why Needed
1	OEE Calculator	Calculates overarching equipment efficiency	Product – Production Monitoring	Availability (%), Performance (%), Quality (%)	OEE (%)	OEE = A × P × Q	Availability, Performance, Quality %	Shows true overall productivity; primary KPI for decision-making
2	Availability Calculator	Measures how much time machine is actually running	Product – Production Monitoring	Run Time (hrs), Planned Time (hrs)	Availability (%)	Availability = Run Time / Planned Time × 100	Utilization	Identifies major idle time losses
3	Performance Calculator	Measures how fast machine produces vs theoretical speed	Product – Production Monitoring	Actual Output (units), Ideal Output Rate (units/hr), Run Time (hrs)	Performance (%)	Performance = (Actual / (Ideal × Run Time)) × 100	OEE	Shows cycle inefficiency bottlenecks
4	Quality % Calculator	Measures good units vs total units produced	Product – Production Monitoring	Total Units, Good Units	Quality (%)	Quality = Good Units / Total Units × 100	FPY	Tracks rejects & scrap contribution
5	Utilization Calculator	Measures use of machine capacity	Product – Production Monitoring	Run Time (hrs), Available Time (hrs)	Utilization (%)	Utilization = Run Time / Available Time × 100	OEE	Exposes hidden idle time shifts
6	Downtime Loss Calculator	Quantifies financial impact of stoppages	Product – Production Monitoring	Downtime (hrs), Output/hr (units/hr), Cost/unit (₹/unit)	Lost Units, Cost Loss	Units = Downtime × Output/hr; Cost = Units × Cost/unit	Production Throughput	Drives urgency for downtime reduction

7	Throughput Gain Calculator	Estimates revenue impact of efficiency improvements	Product – Production Monitoring	Current Output (units/day), Improvement % (%), Price/unit (₹/unit)	Extra Units, Extra Revenue	Extra Units = Output × (Imp%/100); Revenue = Extra Units × Price	OEE, Cycle Time	Highlights increased earning potential
8	Shift Performance Efficiency Calculator	Benchmarks shift productivity	Product – Production Monitoring	Planned Units, Actual Units, Rejected Units	Shift Efficiency %, FPY (%)	Efficiency = Actual/Planned×100; FPY = (Actual–Rejected)/Actual×100	Quality %	Finds lowest performing shifts
9	Bottleneck Identification Calculator	Finds slowest workstation impacting output	Product – Production Monitoring	Cycle Time/Machine (sec), Units per Station	Bottleneck Station ID	Bottleneck = Highest Cycle Time station	Cycle Time Variance	Pinpoints throughput limiters
10	Cycle Time Variance Calculator	Measures inconsistency in manufacturing steps	Product – Production Monitoring	Actual Cycle Time vs Standard (sec)	Cycle Variance (sec/%)	Variance = (Actual – Standard)	Bottleneck	Detects cycle stability issues
11	Module-Level ROI Calculator	Financial ROI specific to Production Monitoring	Product – Production Monitoring	Current OEE %, Improved OEE %, Machine Hour Cost (₹/hr), Daily Hours, #Machines	Annual Savings, ROI %, Payback	Saved Hrs/day = (ΔOEE%×Hrs); Savings = Hrs×Cost×365	OEE	Converts operational gains into monetary justification
12	MTTR Calculator	Measures average repair time per failure	Product – Condition Monitoring System	Total Repair Time (hrs), No. of Breakdowns	MTTR (hrs)	MTTR = Total Repair Time / #Breakdowns	MTBF	Helps evaluate maintenance efficiency

13	MTBF Calculator	Predicts reliability between machine failures	Product – Condition Monitoring System	Total Operating Time (hrs), #Breakdowns	MTBF (hrs)	MTBF = Operating Time / #Breakdowns	MTTR	Shows reliability trend improvement
14	Breakdown Cost Calculator	Calculates expense impact of failure events	Product – Condition Monitoring System	Downtime (hrs), Cost/unit (₹/unit), Output/hr (units/hr)	Cost Loss (₹)	Lost Units = Downtime ×Output/hr; Cost=Lost Units×Cost/unit	Downtime Loss	Highlights financial urgency to prevent failures
15	Predictive Maintenance Savings Calculator	Measures benefit of early fault detection	Product – Condition Monitoring System	Avg Breakdown Cost/event (₹), Predicted Reduction % (%)	Monthly Savings, Annual Savings (₹)	Savings = Cost/event × Reduction % × #events	MTBF	Supports investment in CM sensors & analytics
16	RUL Impact Calculator	Measures life gained due to predictive actions	Product – Condition Monitoring System	Base Life (hrs), Improved Life (hrs)	Life Improvement (hrs/%)	ΔLife = Improved – Base; %Gain=ΔLife/Base×100	MTBF	Shows how prediction increases machine life
17	Spare Parts Optimization Calculator	Prevents overstocking or stockout	Product – Condition Monitoring System	Current Inventory Value (₹), Reduction % (%)	Monthly Cash Release (₹)	Cash Release = Inventory × Reduction %	Breakdown Cost	Reduces maintenance cost by smarter stocking
18	Exception Response Time Savings Calculator	Measures faster issue handling in CM alerts	Product – Condition Monitoring System	Current Response Time (mins), Improved Time (mins), Cost/min (₹), Events/month	Time Saved, Cost Saved	Time = ΔTime×Events; Cost = Time×Cost/min	MTTR	Shows value of automated notifications

19	Module-Level ROI Calculator	Financial ROI for Condition Monitoring	Product – Condition Monitoring System	Annual Breakdown Cost (₹), Reduction % (%), System Cost (₹)	ROI %, Payback Period	ROI%= Savings/System Cost×100; Savings = Breakdown Cost×Reduction%	Breakdown Cost	Justifies predictive maintenance investments
20	Plant-Wide OEE Improvement Calculator	Measures OEE gain impact across multiple lines	**P-CT**	Current OEE (%), Target OEE (%), Machine Hours/day (hrs), Value/hr (₹/hr)	Extra Hrs/day, Value Gain/day, Monthly Gain	Extra Hrs = ΔOEE% × Hrs; Gain = Extra Hrs × Value/hr	OEE	Shows financial upside of improved visibility
21	Exception Response Time Savings Calculator	Financial impact of faster response to events	**P-CT**	Current Response Time (mins), Improved Time (mins), Cost/min (₹), #Events/month	Time Saved, Cost Saved	Time = ΔTime × Events; Cost = Time × Cost/min	MTTR	Highlights need for automated escalations
22	Multi-Line Productivity Gain Calculator	Revenue improvement from all lines combined	**P-CT**	#Lines, Throughput/Line (units/day), Improvement% (%), Price/unit (₹)	Extra Units, Revenue	Extra Units = Lines×Throughput×Imp%; Revenue = Units×Price	Throughput	Helps CXOs measure scalability benefit
23	KPI Deviation Cost Calculator	Cost impact of KPI fluctuations	**P-CT**	Monthly KPI Deviation (units), Cost/unit (₹), Reduction % (%)	Monthly & Annual Savings	Savings = Deviation×Cost×Reduction%	OEE, Quality %	Reduces variability & enforces governance

24	OEE Calculator	Same KPI reused in Control Tower roll-up	**P-CT**	Availability (%), Performance (%), Quality (%)	OEE (%)	OEE = A×P×Q	Availability, Performance, Quality	Key unified KPI for command-view reporting
25	Downtime Loss Calculator	Shows financial urgency of downtime	**P-CT**	Downtime (hrs), Cost/unit (₹), Output/hr (units/hr)	Lost Units, Cost Loss	Units = Hrs×Out/hr; Cost = Units×Cost/unit	MTTR	Drives rapid action on breakdown alerts
26	Defect Rate Calculator	Measures proportion of defective output	**P-CT**	Defective Units, Total Units (units)	Defect %	Defect = Defective/Total×100	FPY, Quality %	Enables early deviation alerts in dashboard
27	Energy Cost Savings Calculator	Identifies high-cost energy spikes	**P-CT**	Monthly kWh, Cost/kWh (₹/kWh), Reduction % (%)	Monthly & Annual Savings	Savings = kWh×Cost ×Reduction %	CO_2 Calculator	Helps justify energy controls & alarms
28	Root Cause Time Savings Calculator	Cost savings from faster investigation	**P-CT**	Avg Time Before vs After (hrs), Cost/hr (₹), #Events/yr	Time Saved, Cost Saved	Time = ΔTime; Cost = Time×Cost/hr	MTTR	Improves exception handling scores
29	Module-Level ROI Calculator	ROI specifically for Control Tower	**P-CT**	Monthly Loss due to visibility gaps (₹), Reduction % (%), System Cost (₹)	ROI %, Payback	Annual Savings = Loss×Reduction%×12; ROI=Save/Cost×100	OEE, KPI	CFO-ready justification

30	Cost of Poor Quality (COPQ) Calculator	Calculates total cost of defective output	**P-QMS**	Rejected Units (units), Cost/unit (₹/unit), Rework Cost/unit (₹/unit), Return Cost (₹)	Total COPQ (₹)	COPQ = (Rejected× Cost) + (Rework×ReworkCost) + Return Cost	Defect Rate	Helps quantify economic impact of quality issues
31	First Pass Yield (FPY) Calculator	Measures quality success on first try	**P-QMS**	Total Units (units), Good Units (units)	FPY (%)	FPY = Good/Total × 100	Quality %	Drives zero-defect culture
32	Defect Rate Calculator	Shows proportion of bad units	**P-QMS**	Defective Units (units), Total Units (units)	Defect Rate (%)	DR = Defective/Total × 100	COPQ	Enables defect alerts & prevention
33	Rejection Reduction Savings Calculator	Shows savings from lower rejection rate	**P-QMS**	Current Rejection% (%), Target Rejection% (%), Monthly Production (units), Cost/unit (₹/unit)	Monthly & Annual Savings (₹)	ReducedUnits = Prod×ΔReject%; Savings=Units×Cost	FPY	Used for improvement justification by quality head
34	Quality Deviation Cost Calculator	Cost of variations in quality KPIs	**P-QMS**	Deviation (units), Cost/unit (₹/unit), Reduction % (%)	Monthly & Annual Savings (₹)	Savings = Dev×Cost× Reduction	COPQ	Reduces variances causing customer complaints
35	Complaint Recall Risk Calculator	Predicts monetary risk of customer recalls	**P-QMS**	Complaint Events/yr (#), Cost/event (₹), Reduction % (%)	Annual Avoided Cost (₹)	Avoided=Events×Cost ×Reduction %	Traceability	Major KPI in OEM audits & risk mitigation

36	Audit Preparation Time Savings Calculator	Measures cost/time savings in audits	**P-QMS**	Hours/Audit (hrs), #Audits/yr, Cost/hr (₹), Automation % (%)	Time Saved (hrs), Money Saved (₹)	Time=Hours×Auto%; Cost=Time×Cost/hr	Traceability	Compliance-heavy industries need digital tracking
37	Module-Level ROI Calculator	ROI specific to QMS implementation	**P-QMS**	Annual Rejection Cost (₹), Improvement% (%), System Cost (₹)	Annual Savings (₹), ROI% (%)	Savings = RejectionCost×Improve%×12; ROI=Save/Cost×100	COPQ, FPY	Shows CXOs clear financial justification
38	Traceability Loss Calculator	Measures cost of missing genealogy data	**P-TRC**	Incidents/month (#), Cost/incident (₹), Reduction % (%)	Monthly Savings (₹), Annual Savings (₹)	Savings = Incidents×Cost×Reduction%	Recall Cost	Prevents disruptions & compliance failures
39	Recall Cost Calculator	Shows financial risk of product recalls	**P-TRC**	Recall Cost/event (₹), #Events/yr, Reduction % (%)	Annual Recall Savings (₹)	Savings = Cost×Events×Reduction%	COPQ	Strong for consumer safety & OEM compliance
40	Audit Preparation Cost Calculator	Savings by automated traceability reports	**P-TRC**	Hours/audit (hrs), #Audits/yr, Cost/hr (₹), Automation% (%)	Time Saved (hrs), Money Saved (₹)	Time=Hours×Auto%; Cost=Time×Cost/hr	QMS	Faster compliance during external audits
41	Root Cause Time Savings Calculator	Speed improvement in defect investigations	**P-TRC**	Avg Investigation Time Before vs After (hrs), Cost/hr (₹), #Events/yr	Time Saved (hrs), Cost Saved (₹)	Time = ΔTime; Cost = Time×Cost/hr	MTTR	Supports quick containment & CAPA action

42	FPY Calculator	Re-used KPI for trace-back context	**P-TRC**	Total Units (units), Good Units (units)	FPY (%)	FPY = Good/Total × 100	Defect Rate	Visibility into where defects originated
43	Module-Level ROI Calculator	ROI of traceability module	**P-TRC**	Loss from Traceability gaps (₹), Recall Cost (₹), Audit Cost (₹), Reduction % (%), System Cost (₹)	Net Annual Savings (₹), ROI % (%)	NetSavings = (Loss+Recall+Audit)×Reduction%; ROI=Save/Cost×100	Recall	Strong ROI driver in regulated industries
44	Energy Cost Savings Calculator	Measures savings from energy optimization	**P-EMS**	Monthly Energy Consumption (kWh), Cost/kWh (₹/kWh), Reduction % (%)	Monthly Savings (₹), Annual Savings (₹)	Savings = kWh×Cost ×Reduction %	CO_2	Main justification for EMS investment
45	Peak Load Penalty Avoidance Calculator	Calculates penalty cost saved by peak control	**P-EMS**	Peak Demand (kVA), Allowed Peak (kVA), Penalty/kVA (₹), Reduction % (%)	Monthly Avoidance (₹), Annual Savings (₹)	Excess = (Peak–Allowed); Avoidance =Excess×Penalty×Reduction%	Energy Savings	Avoids heavy fines in utilities billing
46	Energy per Unit Cost Calculator	Measures energy cost per product	**P-EMS**	Total Energy Cost/month (₹), Total Units Produced (units)	Energy Cost per Unit (₹/unit)	Cost/unit = TotalCost / Units	Energy Savings	Enables cost benchmarking per product or line
47	Carbon Footprint (CO_2) Calculator	Calculates emission impact of consumption	**P-EMS**	Monthly Energy (kWh), CO_2 Factor (kg/kWh)	Monthly CO_2 (kg), Annual CO_2 (kg)	CO_2 = kWh×Factor	Energy Cost	Mandatory metric for ESG compliance

48	Utility Load Efficiency Calculator	Identifies improvement in utility systems	**P-EMS**	Load Consumption Before vs After (kWh), Efficiency Gain % (%)	Monthly Savings (₹), Efficiency %	Savings = ΔkWh×Cost; Efficiency= Gain%	Energy Savings	Reduces waste in chillers/compressors etc.
49	Module-Level ROI Calculator	Financial ROI of EMS deployment	**P-EMS**	Annual Energy Cost (₹), Reduction % (%), System Cost (₹)	ROI %, Payback Period	AnnualSave=Cost×Reduce%×12; ROI=Save/Cost×100	Energy & CO_2	CFO-level justification
50	Cycle Time Savings Calculator	Measures productivity gained by reducing assembly cycle time	**P-AT**	Current Cycle Time (sec), Improved Cycle Time (sec), Units/Shift (units), Shifts/Day (#)	Time Saved (sec), Extra Units/Day (units)	Time = (Current–New)×Units; ExtraUnits = Time/New Cycle	Throughput	Shows speed improvement from digital SOPs
51	Assembly Error Cost Calculator	Financial loss caused by assembly mistakes	**P-AT**	Errors/Month (#), Cost/Error (₹), Reduction % (%)	Monthly & Annual Savings (₹)	Savings = Errors×Cost×Reduction%	COPQ	Quantifies benefits of mistake-proofing
52	Line Balancing Efficiency Calculator	Evaluates workload distribution between stations	**P-AT**	Total Available Cycle Time (sec), Actual Cycle Time Used (sec), Workstations (#)	Line Efficiency (%)	Efficiency = Available/Used×100	Cycle Time Variance	Improves throughput by balancing operators
53	Assembly FPY Calculator	Measures how many assemblies are correct first time	**P-AT**	Total Assemblies (units), Good Assemblies (units)	FPY (%)	FPY = Good/Total ×100	Quality %	Supports zero-defect assembly improvement

54	Andon Escalation Savings Calculator	Savings from faster issue escalation on line	**P-AT**	Current Escalation Time (mins), Improved Time (mins), Cost/min (₹), #Escalations/Month (#)	Time & Cost Saved	Time = ΔTime×Events; Cost=Time ×Cost/min	MTTR	Shows why auto escalations reduce stoppages
55	Module-Level ROI Calculator	ROI specific to Assembly Tracking	**P-AT**	Monthly Error Cost (₹), Monthly Cycle Loss Cost (₹), Reduction % (%), System Cost (₹)	ROI %, Payback(weeks/months)	AnnualSave = (ErrCost+CycleLoss)×Improve%×12; ROI=Save/Cost×100	Cycle Time, FPY	Direct business value for exec decision
56	Access Governance Cost Savings Calculator	Savings from automated user/role provisioning	**P-PLT**	Users (#), Hours/Month spent managing access (hrs), Cost/hr (₹/hr), Automation % (%)	Monthly Savings (₹), Annual Savings (₹)	Savings = Hours × Cost × Automation%	N/A	Digital governance reduces recurring admin cost
57	Integration Cost Avoidance Calculator	Reduces cost of custom ERP/MES/SCADA integrations	**P-PLT**	Cost/System Integration (₹), #Systems (#), Efficiency % (%)	Cost Avoided (₹)	Avoidance = Cost × Systems × Efficiency%	Platform ROI	Major cost center relieved by standard APIs

58	Multi-Plant Scalability Savings Calculator	Savings from single centralized deployment	**P-PLT**	Cost/Plant for independent deployment (₹), Platform Deployment Cost (₹), Maintenance Reduction % (%)	Annual Savings (₹)	Savings = (IndependentCost−PlatformCost) + (PlatformCost×MaintReduce%)	Access Governance	Predictable scaling for enterprise rollouts
59	Data Loss Prevention Value Calculator	Cost avoided by secure, governed data	**P-PLT**	Avg Cost/Data Loss Event (₹), Expected Events/yr (#), Risk Reduction % (%)	Annual Cost Avoided (₹)	Avoided = Cost×Events×Reduction%	MTTR	Critical for cybersecurity justification
60	OEE Roll-Up Calculator	Aggregates OEE from asset→line→plant	**P-PLT**	OEE per Machine (%), Weighting Factors (%)	Consolidated OEE (%)	Weighted OEE = Σ(OEE_i × Weight_i)	OEE, Availability	Gives CXOs single global KPI per site
61	Platform-Level ROI Calculator	Financial ROI of core IIoT platform	**P-PLT**	Annual Fragmentation Cost (₹), Integration Cost (₹), Access Mgmt Cost (₹), Reduction % (%), System Cost (₹)	ROI %, Payback (months)	Savings = (Frag+Integ+Access) ×Reduction%; ROI=Saving/Cost×100	Integration	Executive decision support
62	OEE Calculator	Measures overall efficiency of auto production lines	**I-Auto**	A (%), P (%), Q (%)	OEE (%)	OEE = A×P×Q	Availability, Performance, Quality	OEM suppliers must meet strict delivery rates

63	Downtime Loss Calculator	Cost impact of stoppages in CNC/robot lines	**I-Auto**	Downtime (hrs), Output/hr (units/hr), Cost/unit (₹)	Lost Units, Lost Cost	Units=Hrs×Out/hr; Cost=Units×Cost/unit	MTTR	Critical to avoid OEM line stoppages
64	MTTR Calculator	Avg repair time for welding/CNC failures	**I-Auto**	Total Repair Time (hrs), Breakdowns (#)	MTTR (hrs)	RepairTime/Breakdowns	MTBF	Improves uptime for bottleneck machines
65	MTBF Calculator	Predicts capability of machines before failure	**I-Auto**	Run Time (hrs), Breakdowns (#)	MTBF (hrs)	RunTime/Breakdowns	MTTR	Ensures reliable supply commitments
66	Breakdown Cost Calculator	Loss due to press/weld downtime	**I-Auto**	Downtime (hrs), Cost/unit (₹), Output/hr	Lost Output, Cost Loss	Same as DowntimeLoss	COPQ	Shows link between breakdowns & rejection
67	FPY Calculator	Good parts at machining/assembly first time	**I-Auto**	Total Units, Good Units	FPY (%)	Good/Total×100	Quality %	Improves supplier rating from OEMs
68	Defect Rate Calculator	Proportion of scrap/rejects in machining	**I-Auto**	Defective Units, Total Units	Defect%	Defect/Total×100	FPY	Controls rejection bills from OEM
69	Rejection Reduction Savings	Money saved by fewer part rejections	**I-Auto**	Rejection% Δ, Monthly Output, Cost/unit	Monthly & Annual Savings	SavedUnits = Out×Δ%; Save = Units×Cost	COPQ	Tier-1/Tier-2 profitability improvement

70	Traceability Loss Calculator	Cost from missing part genealogy	**I-Auto**	Incidents (#), Cost/incident, Reduction %	Savings	Inc×Cost×Reduction %	Recall Cost	Mandatory for VIN/part-level tracking
71	Recall Cost Calculator	OEM warranty/recall risk avoidance	**I-Auto**	Cost/event, Events/yr, Reduction %	Avoided Cost	Cost×Events×Reduction%	COPQ	Strong compliance benefit
72	Root Cause Time Savings	Faster defect trace-back during OEM complaints	**I-Auto**	Time Before vs After, Cost/hr	Time & Cost Saved	ΔTime×Cost	FPY	Avoids line stoppage penalties
73	Cycle Time Savings Calculator	Faster APQP assembly lines	**I-Auto**	Cycle Before vs After, Units/Shift, Shifts/Day	Time Saved, Extra Output	Time = ΔCycle×Units	Throughput	Improves takt time for JIT/JIS
74	Assembly Error Cost Calculator	Wrong torque/wiring/weld defects	**I-Auto**	Errors/mo, Cost/error, Reduction %	Savings	Errors×Cost×Reduction%	FPY	Eliminates rework & customer rejections
75	Line Balancing Efficiency	Workload mismatch in assembly cells	**I-Auto**	Available vs Actual Cycle	Efficiency %	Ava/Used×100	Cycle Time Var	Supports lean balancing
76	Plant-Wide OEE Improvement	Multi-line gain across CNC, weld, paint	**I-Auto**	Curr OEE, Target OEE, Hours, Value/hr	Gain Hrs & ₹	ΔOEE×Hrs×Value	OEE	Benchmark across multiple plants
77	Multi-Line Productivity Gain	Additional output from all lines	**I-Auto**	#Lines, Throughput/line, Improvement%, Price/unit	Extra Units, Revenue	Lines×Thru×Imp%	Throughput	Expands supply capacity

78	Energy Cost Savings Calculator	Welding robots/compressors consumption	**I-Auto**	kWh, ₹/kWh, Reduction %	Savings	kWh×Cost ×Reduce%	CO_2	High energy industrial environment
79	CO_2 Footprint Calculator	Environmental reporting to OEM	**I-Auto**	kWh, Factor (kg/kWh)	CO_2 kg	kWh×Factor	Energy	Supports sustainability compliance
80	Module-Level ROI Calculator	Combined financial benefit from auto digitization	**I-Auto**	Annual Losses, Reduction %, System Cost	ROI%, Payback	Savings=Loss×Reduction%; ROI=Save/Cost×100	All above	CFO visibility for adoption
81	OEE Calculator	Measures line performance for high-speed food lines	**I-FMCG**	A(%),P(%), Q(%)	OEE (%)	A×P×Q	Availability, Quality	Major KPI for bottleneck packaging lines
82	Throughput Gain Calculator	Added packs/bottles from speed improvements	**I-FMCG**	Output/day (units), Improve% (%), Price/unit (₹)	Extra Units, Revenue	Extra = Out×Imp%; Rev=Units ×Price	Cycle Time	Critical in high-speed lines
83	Wastage Cost Calculator	Cost of product wasted/spoiled	**I-FMCG**	Wastage (kg/l), Cost/kg (₹), Reduce% (%)	Daily/Monthly Savings	Savings = Waste×Cost×Reduce%	Quality %	Food spoilage is major profit loss
84	FPY Calculator	Good packs produced without rework	**I-FMCG**	Total Units, Good Units	FPY (%)	Good/Total ×100	Defect Rate	Helps minimize customer complaints
85	Quality Deviation Cost	Yield inconsistency impact	**I-FMCG**	Deviations (#), Cost/unit, Reduce%	Monthly & Annual Savings	Dev×Cost× Red%	FPY	Prevents flavour/weight variations

86	Downtime Loss Calculator	Packaging flow stoppage losses	**I-FMCG**	Downtime hrs, Output/hr, Cost/unit	Lost units (units), Lost cost (₹)	Hrs×Out×Cost	MTTR	Downtime on fillers → big losses
87	Energy Cost Savings	Chillers, compressors, ovens usage optimization	**I-FMCG**	kWh, ₹/kWh, Reduce%	Monthly & Annual Savings	kWh×Cost ×Red%	CO_2	Utility-heavy food plants
88	CO_2 Calculator	Emissions reporting	**I-FMCG**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Sustainability audits, export compliance
89	Audit Preparation Cost	HACCP, GMP, FSSAI audits readiness	**I-FMCG**	Hours, #audits, Cost/hr, Auto%	Time & Cost Saved	Hours×Auto%×Cost	QMS	Reduces compliance workload
90	Cycle Time Variance	Inconsistency in packaging lines	**I-FMCG**	Cycle vs Std(sec)	Variance (sec)	Cycle–Std	Line Balancing	Exposes filling/packing inefficiency
91	Module-Level ROI Calculator	ROI for FMCG digitization	**I-FMCG**	Waste Cost, Downtime Cost, Energy Cost, Improve%, System cost	ROI%, Payback	ΣLoss×Imp% vs Cost	OEE, Energy	CFO scorecard for justification
92	FPY Calculator	Batch accuracy & first-time-right	**I-Pharma**	Total Batches, Good Batches	FPY (%)	Good/Total ×100	Quality %	Essential for batch consistency
93	COPQ Calculator	Rework, scrap & lab test cost	**I-Pharma**	Reject Units, Cost/unit, Rework Cost	Total COPQ	Same as before	Defect Rate	Controls cost of failed batches

94	Audit Preparation Cost Calculator	Validation, GMP, FDA audit readiness	**I-Pharma**	Hours, Cost/hr, Auto%	Time & Cost Saved	Hours×Auto%×Cost	Traceability	Mandatory compliance requirement
95	Traceability Loss Calculator	Genealogy gaps → regulatory risk	**I-Pharma**	Incidents, Cost/incident, Reduce%	Savings	Inc×Cost×Reduce%	Recall	Required for serialisation
96	Recall Cost Calculator	Drug recalls risk cost	**I-Pharma**	Cost/event, Events, Reduce%	Avoided Cost	Cost×Events×Reduce%	FPY	Life & reputation critical
97	Root Cause Time Savings	Investigation time reduction	**I-Pharma**	Before vs After, Cost/hr	Savings	ΔTime×Cost	MTTR	Faster CAPA improves approvals
98	Energy Cost Savings	HVAC, cleanrooms power load	**I-Pharma**	kWh, Cost, Reduce%	Savings	kWh×Cost×Red%	CO_2	Energy-compliance must
99	CO_2 Calculator	Emissions tracking	**I-Pharma**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	ESG proof for regulators
100	Module-Level ROI Calculator	ROI for pharma digitization	**I-Pharma**	Losses (₹), Improve%, System Cost	ROI%, Payback	Loss×Imp%×12 vs Cost	All	Multi-dept compliance value
101	OEE Calculator	Efficiency of weaving/processing lines	**I-Text**	A,P,Q	OEE (%)	A×P×Q	Utilization	Drives yarn-to-fabric throughput
102	Downtime Loss Calculator	Loom stoppage, breaks, delays impact	**I-Text**	Downtime hrs, Output/hr, Cost/unit	Lost cost	Hrs×Out×Cost	MTTR	Fabric delays → high loss
103	FPY Calculator	Quality of weaving, dyeing success	**I-Text**	Total units, Good units	FPY (%)	Good/Total×100	Quality%	Eliminates reweaving/repairs

104	Defect Rate Calculator	Holes, stains, color deviation tracking	**I-Text**	Defective, Total	%	Def/Total×100	COPQ	Key for inspection analytics
105	Energy Cost Savings	Boilers, spindles, dyehouse loads	**I-Text**	kWh, Cost/kWh, Reduce%	Savings	kWh×Cost×Reduce%	CO_2	Textile is highly energy intensive
106	CO_2 Calculator	Dyehouse emission reporting	**I-Text**	kWh, Factor	CO_2	kWh×Factor	Energy	Sustainability initiatives
107	Cycle Time Variance	Stitching line consistency	**I-Text**	Cycle vs Std	Variance	Cycle–Std	Bottleneck	Supports lean cell improvements
108	Module-Level ROI Calculator	Textile digitization ROI	**I-Text**	Loss, Improve%, Cost	ROI	Loss×Imp%×12 vs Cost	OEE	Cost-sensitive mass production
109	OEE Calculator	SMT & assembly line efficiency	**I-Elec**	A(%), P(%), Q(%)	OEE (%)	A×P×Q	Availability, Quality	Critical to global electronics KPIs
110	FPY Calculator	Good units from pick-n-place line	**I-Elec**	Total, Good	FPY (%)	Good/Total×100	Quality %	Direct ties to customer acceptance
111	Defect Rate Calculator	Defects in PCBA testing, soldering	**I-Elec**	Defective, Total	%	Def/Total×100	FPY	Electronics rework cost is high
112	Rejection Reduction Savings	Avoid PCBA scrap cost	**I-Elec**	Reject%Δ, Output, Cost/unit	Savings	Units×Cost	COPQ	Improves profitability under tight margins
113	Traceability Loss Calculator	Missing PCBA/serial tracking	**I-Elec**	Incidents, Cost, Reduce%	Savings	Inc×Cost×Red%	Recall Cost	Regulatory and safety requirement

114	Root Cause Time Savings	Faster test/debug resolution	**I-Elec**	Before/After hr, Cost/hr	Time & Cost	ΔTime×Cost	FPY	Faster issue isolation reduces downtime
115	Energy Cost Savings	SMD ovens, compressors, HVAC	**I-Elec**	kWh, Cost, Reduce%	Savings	kWh×Cost×Red%	CO_2	Electronics factories high energy density
116	CO_2 Footprint Calculator	ESG reporting	**I-Elec**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Needed for export audits
117	Module-Level ROI Calculator	ROI for electronics digitization	**I-Elec**	Losses, Improve%, Cost	ROI%	Loss×Imp%×12 vs Cost	OEE, Traceability	CFO decision-metric
118	OEE Calculator	Efficiency of molding cycle	**I-Plas**	A,P,Q	OEE	A×P×Q	Utilization	Crucial for cavity-cycle optimization
119	Cycle Time Variance	Molding cycle consistency	**I-Plas**	Cycle vs Std (sec)	Variance	Cyc–Std	Bottleneck	Improves throughput on molding presses
120	Downtime Loss Calculator	Mold cooling or press breakdown loss	**I-Plas**	Hrs, Out/hr, Cost/unit	Lost output & cost	Hrs×Out×Cost	MTTR	Minimizes mould-change delays
121	Defect Rate Calculator	Flashing, burn marks, short molding	**I-Plas**	Defect, Total	%	Def/Total×100	COPQ	Quality driven customer penalties
122	FPY Calculator	First-shot-right	**I-Plas**	Total, Good	FPY%	Good/Total×100	Quality%	Reduces scrap of costly polymers

123	Energy Cost Savings	Heaters, chillers, dryers	**I-Plas**	kWh, Cost, Reduce%	Savings	kWh×Cost ×Red%	CO_2	Injection molding is energy intensive
124	Utility Load Efficiency	Compressed air and cooling efficiency	**I-Plas**	Load Before/After, Gain%	Savings	ΔkWh×Cost	Energy Savings	Supports stable machine operation
125	Module-Level ROI Calculator	ROI for plastics digitization	**I-Plas**	Loss, Improve%, Cost	ROI%	Loss×Imp% ×12 vs Cost	OEE	High-volume manufacturers profitability
126	MTTR Calculator	Repair time in heavy machines	**I-Heavy**	Total Repair Hrs, Breakdowns	MTTR	Hours/Event	MTBF	Heavy tools → long downtime
127	MTBF Calculator	Reliability of furnaces, presses	**I-Heavy**	Run Hrs, Events	MTBF	Hrs/Breakdown	MTTR	Ensures continuous forging ops
128	Breakdown Cost Calculator	Severe loss when furnaces stop	**I-Heavy**	Hrs, Out/hr, Cost/unit	Lost cost	Hrs×Out×Cost	Downtime	Crucial for ROI justification
129	Energy Cost Savings	Furnaces, compressors, pumps	**I-Heavy**	kWh, Cost, Reduce%	Savings	kWh×Cost ×Red%	CO_2	Highest energy intensity
130	CO_2 Calculator	Emission reporting for sustainability	**I-Heavy**	kWh, Factor	CO_2	kWh×Factor	Energy Savings	For ESG + environment boards
131	FPY Calculator	Scrap reduction in castings	**I-Heavy**	Total, Good	FPY%	Good/Total ×100	Defect Rate	Eliminates costly steel scrap
132	Defect Rate Calculator	Casting defects like porosity	**I-Heavy**	Defect, Total	%	Def/Total×100	COPQ	Essential in metallurgical control

133	Module-Level ROI Calculator	ROI for heavy engineering digitization	**I-Heavy**	Loss×, Improve%, Cost	ROI%	Loss×Imp%×12 / Cost	Breakdown	CFO-grade visibility
134	OEE Calculator	Efficiency of machine test beds	**I-OEM**	A,P,Q	OEE%	A×P×Q	Cycle Time	Improves delivery timelines
135	Throughput Gain Calculator	More machines shipped/month	**I-OEM**	Output/day, Improve%, Price/unit	Extra Units & Revenue	Out×Imp%	Availability	High business growth driver
136	MTTR Calculator	Repair time during FAT/SAT	**I-OEM**	Repair Hrs, Events	MTTR	Hrs/Event	MTBF	Improves commissioning speed
137	MTBF Calculator	Reliability during machine testing	**I-OEM**	Run Hrs, Events	MTBF	Hrs/Event	MTTR	Strong KPI for machine quality
138	FPY Calculator	First-time-right at build-validation	**I-OEM**	Total Units, Good Units	FPY%	Good/Total×100	Defect Rate	Reduces production delays
139	Defect Rate Calculator	Build errors detection	**I-OEM**	Defect, Total	%	Def/Total×100	FPY	Strong impact on on-time delivery
140	Traceability Loss Calculator	Assembly genealogy trace	**I-OEM**	Incidents, Cost, Reduce%	Savings	Inc×Cost×Red%	Recall	Required by enterprise buyers
141	Root Cause Time Savings	Faster troubleshooting	**I-OEM**	Before/After hrs, Cost/hr	Cost Saved	ΔTime×Cost	MTTR	Reduces commissioning delays
142	Module-Level ROI Calculator	ROI for OEM digitization	**I-OEM**	Delay Losses, Improve%, Cost	ROI%	ΣLoss×Imp%×12 / Cost	OEE	Improves order-to-cash cycle

143	OEE Calculator	Efficiency of welding & cutting	**I-Fab**	A,P,Q	OEE%	A×P×Q	Performance	Ensures production schedule adherence
144	Downtime Loss Calculator	Loss from CNC/laser machine stops	**I-Fab**	Hrs, Out/hr, Cost/unit	Lost Cost	Hrs×Out×Cost	MTTR	High hourly cost per machine
145	FPY Calculator	Good welds & bends first time	**I-Fab**	Total, Good	FPY%	Good/Total×100	Defect Rate	Ensures minimal rework/wastage
146	Defect Rate Calculator	Weld porosity, mis-cuts, burrs	**I-Fab**	Defect, Total	%	Def/Total×100	COPQ	Drives inspection excellence
147	Energy Cost Savings	Cutting, forming, compressors	**I-Fab**	kWh, Cost, Reduce%	Savings	kWh×Cost×Red%	CO_2	Fabrication shops = high energy
148	CO_2 Calculator	Emission from heavy equipment	**I-Fab**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Helps sustainability reports
149	Root Cause Time Savings	Faster error identification	**I-Fab**	Time Δ, Cost/hr	Savings	ΔTime×Cost	MTTR	Reduces line downtime impact
150	Module-Level ROI Calculator	ROI of fab digitization	**I-Fab**	Loss, Improve%, Cost	ROI%	ΣLoss×Imp%×12/Cost	OEE	Strong CFO decision driver
151	FPY Calculator	Zero-defect production mandate	**I-Aero**	Total, Good	FPY%	Good/Total×100	Defect Rate	Mandatory for A&D supply chain
152	Defect Rate Calculator	Part scrappage: tolerance critical	**I-Aero**	Defect, Total	%	Def/Total×100	FPY	Critical cost exposure

153	COPQ Calculator	Cost of rework/precision scrap	**I-Aero**	Rejects×Cost + Rework cost	COPQ	—	FPY	Sensitive pricing margins
154	Traceability Loss Calculator	Missing part serial or batch link	**I-Aero**	Incidents, Cost, Reduce%	Savings	Inc×Cost×Red%	Recall	Compliance to AS9100 standards
155	Recall Cost Calculator	Defense recall/warranty risk	**I-Aero**	Cost/event, Events, Red%	Avoided Cost	Cost×Events×Red%	COPQ	National-security supply chain
156	Audit Preparation Cost	Aerospace audits readiness	**I-Aero**	Hrs, Cost/hr, Auto%	Savings	Hrs×Cost×Auto%	QMS	Heavy compliance audit load
157	Root Cause Time Savings	Rapid NC/Deviation closure	**I-Aero**	Time Δ, Cost/hr	Savings	ΔTime×Cost	MTTR	CAPA closure KPIs
158	Energy Cost Savings	Facility energy usage	**I-Aero**	kWh, ₹/kWh, Red%	Savings	kWh×Cost×Red%	CO_2	High rate heat-treatment
159	Module-Level ROI Calculator	ROI for A&D compliance	**I-Aero**	Losses, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	FPY, Traceability	Critical compliance value
160	FPY Calculator	Good units first time in precision devices	**I-Med**	Total, Good	FPY%	Good/Total×100	Quality%	Prevents defective medical parts
161	Defect Rate Calculator	Scrap & defects in med assemblies	**I-Med**	Defect, Total	%	Def/Total×100	FPY	Strict safety standards
162	COPQ Calculator	Cost of rework & scrappage	**I-Med**	Rejects, Cost/unit, Rework Cost	COPQ	Reject×Cost+ReworkCost	Defect Rate	Quality = brand reputation
163	Traceability Loss Calculator	Missing UDI/serial link impact	**I-Med**	Incidents, Cost, Red%	Savings	Inc×Cost×Red%	Recall Cost	Required for UDI compliance

164	Recall Cost Calculator	Recall risk cost	**I-Med**	Cost/event, Events, Red%	Avoided Cost	Cost×Events×Red%	COPQ	Healthcare safety requirement
165	Audit Prep Cost Calculator	FDA, ISO audits digitization	**I-Med**	Hrs, Cost/hr, Auto%	Savings	Hrs×Cost×Auto%	QMS	Regulatory compliance priority
166	MTTR Calculator	Maintenance during device testing	**I-Med**	Repair hrs, Events	MTTR	Hrs/Event	MTBF	Less delay in certification
167	Module-Level ROI Calculator	ROI for medical digitization	**I-Med**	Loss, Improve%, System Cost	ROI%	Loss×Imp%×12/Cost	FPY, Traceability	Compliance-driven ROI
168	OEE Calculator	Line efficiency for appliances	**I-CD**	A,P,Q	OEE%	A×P×Q	Availability	Boost output in assembly lines
169	FPY Calculator	Quality of high volume assemblies	**I-CD**	Total, Good	FPY%	Good/Total×100	Quality%	Reduces warranty defects
170	Defect Rate Calculator	Panel cracks, paint issues	**I-CD**	Defect, Total	%	Def/Total×100	COPQ	Protects brand reputation
171	Assembly Error Cost Calculator	Wrong fitments in assembly	**I-CD**	Errors/mo, Cost/error, Red%	Savings	Errors×Cost×Red%	FPY	High-volume loss driver
172	Cycle Time Savings	Faster takt in conveyor lines	**I-CD**	ΔCycle, Units/Shift	Extra Units	Extra = ΔCycle×Units	Throughput	Increases output w/o new machines
173	Energy Cost Savings	Compressor, HVAC consumption	**I-CD**	kWh, Cost, Red%	Savings	kWh×Cost×Red%	CO_2	Appliances test labs power-intensive
174	CO_2 Calculator	Emission monitoring	**I-CD**	kWh, Factor	CO_2	kWh×Factor	Energy	ESG focus for consumer brands

175	Module-Level ROI Calculator	ROI for digitizing CD plants	**I-CD**	Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	OEE, FPY	Strong CFO sell-point
176	Downtime Loss Calculator	Curing press stoppage losses	**I-Rub**	Hrs, Out/hr, Cost	Lost Cost	Hrs×Out×Cost	MTTR	Downtime kills throughput
177	FPY Calculator	First-pass yield for tyres	**I-Rub**	Total, Good	FPY%	Good/Total×100	Quality%	Reduces scrap & cost per tyre
178	Defect Rate Calculator	Tyre defects like bubble, runout	**I-Rub**	Defect, Total	%	Def/Total×100	COPQ	Improves OEM acceptance rate
179	Energy Cost Savings	Boiler & curing energy	**I-Rub**	kWh, Cost, Red%	Savings	kWh×Cost×Red%	CO_2	Major cost center
180	CO_2 Calculator	GHG emission reporting	**I-Rub**	kWh, Factor	CO_2	kWh×Factor	Energy	Sustainability & compliance
181	Module-Level ROI Calculator	ROI for tyre digitization	**I-Rub**	Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	OEE	Better profitability and compliance
182	OEE Calculator	Rolling/casting efficiency	**I-Steel**	A,P,Q	OEE%	A×P×Q	Utilization	Helps optimize steel mill throughput
183	MTTR Calculator	Maintenance time of mill drives	**I-Steel**	Repair hrs, Events	MTTR	hrs/event	MTBF	Heavy downtime cost
184	MTBF Calculator	Machine reliability metric	**I-Steel**	Run hrs, Events	MTBF	hrs/event	MTTR	Prevents major equipment outages
185	Downtime Loss Calculator	High hourly loss in mills	**I-Steel**	Hrs, Out/hr, Cost/unit	Lost Cost	Hrs×Out×Cost	Breakdown	Production loss is extremely costly

186	Energy Cost Savings	Furnace & rolling energy	**I-Steel**	kWh, Cost, Reduce%	Savings	kWh×Cost ×Red%	CO_2	Energy is major cost driver
187	CO_2 Calculator	Emissions tracking for ESG	**I-Steel**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Mandatory CSR reporting
188	Root Cause Time Savings	Reduce investigation delay	**I-Steel**	ΔTime, Cost/hr	Savings	ΔTime×Cost	MTTR	Faster RCA avoids production delays
189	Module ROI Calculator	ROI of digitizing steel processes	**I-Steel**	Loss, Imp%, Cost	ROI%	Loss×Imp% ×12/Cost	OEE, Energy	High cost, high ROI payoff
190	OEE Calculator	Reels & packaging line efficiency	**I-Pack**	A,P,Q	OEE%	A×P×Q	Throughput	Optimizes paper roll productivity
191	FPY Calculator	Good sheets without tear/wrinkle	**I-Pack**	Total, Good	FPY%	Good/Total ×100	Defect Rate	Reduces trim waste
192	Defect Rate Calculator	Paper tear, mis-print issues	**I-Pack**	Defect, Total	%	Def/Total×100	COPQ	Improves customer acceptance
193	Wastage Cost Calculator	Trim/waste cost measurement	**I-Pack**	Waste (kg), Cost/kg, Red%	Savings	Waste×Cost×Red%	FPY	High waste = major cost center
194	Energy Cost Savings	Pulp processing energy	**I-Pack**	kWh, Cost, Red%	Savings	kWh×Cost ×Red%	CO_2	Heavy power usage
195	CO_2 Calculator	Emission reporting	**I-Pack**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Way for sustainability proof
196	Module ROI Calculator	ROI for packaging plants digitization	**I-Pack**	Loss, Imp%, Cost	ROI%	Loss×Imp% ×12/Cost	OEE, Waste	Justifies automation upgrade cost

197	OEE Calculator	Efficiency optimization in large-item build	**I-IEq**	A,P,Q	OEE%	A×P×Q	Performance	Reduce delays + rework
198	MTTR Calculator	Assembly breakdown recovery	**I-IEq**	Repair hrs, Events	MTTR	hrs/event	MTBF	Quick fault recovery crucial
199	FPY Calculator	Build quality of custom machines	**I-IEq**	Total, Good	FPY%	Good/Total ×100	Defect Rate	First time right = project profit
200	Defect Rate Calculator	Mfg defects in capital goods	**I-IEq**	Defect, Total	%	Def/Total×100	FPY	Reduces rework & customer claims
201	Traceability Loss Calculator	Missing component genealogy	**I-IEq**	Incidents, Cost, Red%	Savings	Inc×Cost×Red%	Recall	Mandatory for after-sales support
202	Audit Prep Cost	Compliance audit savings	**I-IEq**	Hrs, Cost/hr, Auto%	Savings	Hrs×Cost×Auto%	QMS	Industrial safety compliance
203	Module ROI Calculator	ROI for industrial digitization	**I-IEq**	Loss, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	OEE	Needed for CFO approvals
204	FPY Calculator	Batch success rate (first-time-right)	**I-Chem**	Total Batches, Good Batches	FPY%	Good/Total ×100	Quality%	Mandatory for batch consistency
205	COPQ Calculator	Loss from re-blending, waste	**I-Chem**	Reject Qty, Cost/unit, Rework Cost	COPQ (₹)	Reject×Cost + ReworkCost	Defect Rate	Paint waste = huge cost center
206	Defect Rate Calculator	Quality variation, shade mismatch	**I-Chem**	Defective, Total	%	Def/Total×100	FPY	Zero-defect consistency focus
207	Downtime Loss Calculator	Stoppage loss in reactors/mixers	**I-Chem**	Downtime (hrs), Output/hr, Cost/unit	Lost Cost	Hrs×Out×Cost	MTTR	Chemical reaction delays costly

208	Energy Cost Savings	Boilers, pumps, mixers energy	**I-Chem**	kWh, Cost, Reduce%	Savings	kWh×Cost ×Red%	CO_2	Energy heavy batch processes
209	CO_2 Calculator	Emission data for compliance	**I-Chem**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Regulatory compliance required
210	Audit Prep Cost	EHS, ISO, quality audits	**I-Chem**	Hours, Cost/hr, Auto%	Cost Saved	Hrs×Cost× Auto%	QMS	Compliance-driven benefit
211	Module-Level ROI Calculator	ROI of chemical plant digitization	**I-Chem**	Loss, Improve%, Cost	ROI%, Payback	Loss×Imp% ×12/Cost	FPY	Strong CFO selling point
212	OEE Calculator	High-precision cell assembly performance	**I-EV**	A,P,Q	OEE%	A×P×Q	Availability	Used to meet EV demand growth
213	FPY Calculator	Good cells/modules first pass	**I-EV**	Total, Good	FPY%	Good/Total ×100	Quality%	Reduces rework of costly cells
214	Defect Rate Calculator	EV cell defect tracking	**I-EV**	Defect, Total	%	Def/Total×100	FPY	Essential for safety & performance
215	Traceability Loss Calculator	Battery genealogy risk management	**I-EV**	Incidents, Cost, Reduce%	Savings	Inc×Cost× Red%	Recall	Needed for battery certification
216	Recall Cost Calculator	Warranty/recall event cost	**I-EV**	Cost/event, Events, Red%	Avoided Cost	Cost×Events×Red%	COPQ	Safety critical component
217	Root Cause Time Savings	Faster failure detection	**I-EV**	Time Δ, Cost/hr	Savings (₹)	ΔTime×Cost	MTTR	Speeds up corrective actions

218	Energy Cost Savings	Cleanrooms, drying, ovens	**I-EV**	kWh, Cost/kWh, Reduce%	Savings	kWh×Cost ×Red%	CO_2	Very high energy load industry
219	CO_2 Calculator	Sustainability score reporting	**I-EV**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	For green compliance claims
220	Module-Level ROI Calculator	ROI for EV digitization	**I-EV**	Losses, Improve%, System Cost	ROI%	Loss×Imp% ×12/Cost	OEE, Trace	Justifies digital transformation
221	OEE Calculator	Efficiency of discrete machines	**M-Disc**	A,P,Q	OEE%	A×P×Q	Availability	Key KPI for discrete throughput
222	MTTR Calculator	Repair time for machine stops	**M-Disc**	Hrs, Events	MTTR	Hrs/Event	MTBF	Cuts breakdown delays
223	MTBF Calculator	Reliability indicator	**M-Disc**	Run Hrs, Events	MTBF	Run/Event	MTTR	Supports predictive decisions
224	Downtime Loss Calculator	Cost of idle machines	**M-Disc**	Hrs, Out/hr, Cost/unit	Lost Cost	Hrs×Out×Cost	Breakdown	Highlights money leakage
225	FPY Calculator	Good parts first-time-right	**M-Disc**	Total, Good	FPY%	Good/Total ×100	Quality%	Drives scrap reduction
226	Defect Rate Calculator	Scrap tracking	**M-Disc**	Defect, Total	%	Def/Total×100	COPQ	Cost & quality control
227	Cycle Time Variance	Process stability	**M-Disc**	Cycle, Std	Variance	Cycle–Std	Bottleneck	Balances workstations
228	Energy Cost Savings	Machine power usage	**M-Disc**	kWh, Cost, Red%	Savings	kWh×Cost ×Red%	CO_2	Operational efficiency
229	CO_2 Calculator	Emissions per energy use	**M-Disc**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	ESG compliance

230	Module Level ROI	ROI for discrete digitization	**M-Disc**	Loss, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	All above	CFO decision tool
231	OEE Calculator	Continuous line efficiency	**M-Proc**	A,P,Q	OEE%	A×P×Q	Availability	Optimizes throughput
232	MTTR Calculator	Critical maintenance time	**M-Proc**	Repair hrs, Events	MTTR	Hr/Event	MTBF	Minimizes stoppage
233	MTBF Calculator	Continuous line reliability	**M-Proc**	Run hrs, Events	MTBF	Run/Event	MTTR	Prevents shutdown
234	Downtime Loss Calculator	Cost of production drop	**M-Proc**	Hrs, Output/hr, Cost/unit	Lost cost	Hrs×Out×Cost	Breakdown	Cost per hour is huge
235	Energy Cost Savings	Furnace, kiln power	**M-Proc**	kWh, Cost, Red%	Savings	kWh×Cost×Red%	CO_2	Reduces heavy utility expenses
236	CO_2 Calculator	Carbon emission check	**M-Proc**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Mandatory regulations
237	Root Cause Time Savings	Faster deviation resolution	**M-Proc**	Time Δ, Cost/hr	Savings	ΔTime×Cost	MTTR	Prevents quality risk
238	Module Level ROI Calculator	ROI for continuous ops	**M-Proc**	Loss, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	All	Strong justification metric
239	FPY Calculator	Batch yield success	**M-Batch**	Total, Good	FPY%	Good/Total×100	Quality%	Ensures process consistency
240	COPQ Calculator	Failure batch cost	**M-Batch**	Reject, Cost/unit, Rework Cost	COPQ	Reject×Cost+Rework	FPY	Huge loss on failed batches
241	Audit Prep Cost Calculator	GMP compliance automation	**M-Batch**	Hours, Cost/hr, Auto%	Savings	Hrs×Cost×Auto%	Traceability	Compliance-heavy process

242	Traceability Loss Calculator	Genealogy gap cost	**M-Batch**	Inc, Cost, Red%	Savings	Inc×Cost×Red%	Recall	Mandatory for pharma/food
243	Recall Cost Calculator	Recall risk & prevention	**M-Batch**	Cost/event, Events, Red%	Avoided Cost	Cost×Events×Red%	COPQ	Compliance & safety
244	Energy Cost Savings	Boilers, mixers energy	**M-Batch**	kWh, Cost, Red%	Savings	kWh×Cost×Red%	CO_2	High energy consumption
245	CO_2 Calculator	ESG metrics	**M-Batch**	kWh, Factor	CO_2 (kg)	kWh×Factor	Energy	Required for sustainability
246	Module Level ROI Calculator	ROI in batch ops	**M-Batch**	Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	FPY, Trace	CFO-level approval
247	OEE Calculator	Line efficiency & takt adherence	**M-Asm**	A,P,Q	OEE%	A×P×Q	Performance	Core assembly metric
248	FPY Calculator	Good assemblies without rework	**M-Asm**	Total, Good	FPY%	Good/Total×100	Defect Rate	Critical for throughput
249	Defect Rate Calculator	Assembly defects rate	**M-Asm**	Defect, Total	%	Def/Total×100	COPQ	Prevents rework cost
250	Cycle Time Savings Calculator	Takt time improvements	**M-Asm**	ΔCycle, Units/Shift	Extra Output	ΔCycle×Units	Throughput	More output same resources
251	Assembly Error Cost Calculator	Wrong part placement cost	**M-Asm**	Errors/mo, Cost/error, Red%	Savings	Err×Cost×Red%	FPY	Strong Poka-Yoke need
252	Line Balancing Efficiency	Workload distribution metric	**M-Asm**	Ava/Actual Cycle	Efficiency %	Ava/Actual×100	Cycle Variance	Reduces bottlenecks

253	Andon Escalation Savings	Faster problem escalation	**M-Asm**	ΔTime, Cost/min, Events	Savings	ΔTime×Cost×Events	MTTR	Minimum stoppage period
254	Energy Cost Savings	Robot, conveyor energy	**M-Asm**	kWh, Cost, Red%	Savings	kWh×Cost ×Red%	CO_2	Reduce utility overhead
255	CO_2 Calculator	Emissions per line	**M-Asm**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Sustainability reporting
256	Module-Level ROI Calculator	Assembly digitization ROI	**M-Asm**	Loss, Imp%, Cost	ROI%	Loss×Imp% ×12/Cost	OEE	Key CFO justification
257	Energy Cost Savings Calculator	Cost saved in utilities	**M-Util**	kWh, Cost, Red%	Savings	kWh×Cost ×Red%	CO_2	Main KPI for this type
258	Utility Load Efficiency Calculator	Utility consumption improvement	**M-Util**	Load Before/After, Gain%	Savings	ΔkWh×Cost	Energy Cost	Reduces overall plant consumption
259	Peak Load Penalty Avoidance	Avoid utility penalty cost	**M-Util**	Peak, Allowed, Penalty/kVA, Red%	Savings	(Peak–Allowed)×Penalty×Red%	Energy Saving	Avoids penalty billing
260	Energy per Unit Cost Calculator	Utility cost per output	**M-Util**	Total Eng Cost, Units	Cost/unit	Cost/Units	Energy Saving	Benchmark performance
261	CO_2 Calculator	Carbon emission indicator	**M-Util**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Supports ESG targets
262	MTTR Calculator	Repair time in utilities	**M-Util**	Repair hrs, Events	MTTR	Hrs/Event	MTBF	Quick fix to avoid outages
263	MTBF Calculator	Reliability of utility assets	**M-Util**	Run hrs, Events	MTBF	Run/Event	MTTR	Stability & uptime

264	Downtime Loss Calculator	Loss from utility failure	**M-Util**	Hrs×Cost×Output	Lost Cost	Hrs×Out×Cost	Breakdown	Shows risk to production supply
265	Module-Level ROI Calculator	ROI for energy optimization	**M-Util**	Energy Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	All Energy	CFO-level decision value
266	MTTR Calculator	Measures repair speed to reduce downtime	**PR-DT**	Repair hrs, Breakdowns	MTTR	Repair/Events	MTBF	Direct downtime reduction metric
267	MTBF Calculator	Measures time between failures	**PR-DT**	Run hrs, Events	MTBF	Run/Events	MTTR	Better reliability → less downtime
268	Downtime Loss Calculator	Shows financial impact of downtime	**PR-DT**	Hrs, Out/hr, Cost/unit	Lost ₹	Hrs×Out×Cost	OEE	CFO urgency to reduce downtime
269	Breakdown Cost Calculator	Cost per breakdown incident	**PR-DT**	Same as above	Lost ₹	Same calc	MTTR	Prioritizes critical failure fixes
270	Exception Response Time Savings	Cost saved by quicker action	**PR-DT**	ΔTime, Cost/min, Events	Saved ₹	ΔTime×Cost×Events	MTTR	Justifies alerts/escalation
271	Root Cause Time Savings	Faster RCA reduces stop time	**PR-DT**	ΔTime, Cost/hr	Saved ₹	ΔTime×Cost	MTBF	Supports preventive decisions
272	Plant-wide OEE Gain	Uptime improvement effect	**PR-DT**	Curr OEE, Target OEE	OEE Gain	ΔOEE×Hrs×Value	OEE	Exec-level visibility
273	ROI Calculator	Financial impact of downtime reduction	**PR-DT**	Loss ₹, Reduce%, Cost ₹	ROI	Loss×Red%×12/Cost	All Above	Converts value to investment case

274	FPY Calculator	Measures first-time-right	**PR-SR**	Total, Good	FPY%	Good/Total ×100	Quality %	Scrap cost driver
275	Defect Rate Calculator	Reject units proportion	**PR-SR**	Defect, Total	%	Def/Total×100	COPQ	Scrap source detection
276	COPQ Calculator	Cost of poor quality	**PR-SR**	Reject×Cost + ReworkCost	₹	Reject×Cost+Rework	Defect%	CFO tracks COPQ strongly
277	Rejection Reduction Savings	₹ saved by scrap reduction	**PR-SR**	ΔReject%, Output, Cost/unit	Saved ₹	Δ%×Out×Cost	FPY	Shows business gain
278	Complaint Recall Risk	Recalls due to poor quality	**PR-SR**	Events, Cost/event, Red%	Avoided ₹	Cost×Events×Red%	Traceability	Major risk for OEMs
279	Root Cause Time Savings	Speed of fixing quality issues	**PR-SR**	ΔTime×Cost/hr	Saved ₹	ΔTime×Cost	RCA	Drives digital CAPA demand
280	ROI Calculator	ROI specific to quality	**PR-SR**	Loss ₹, Improve%, SysCost	ROI%	Loss×Imp%×12/Cost	FPY	Solid justification model
281	OEE Calculator	Unified KPI visibility	**PR-LV**	A,P,Q	OEE	A×P×Q	Availability	Seeing losses drives action
282	Availability Calculator	Real-time runtime tracking	**PR-LV**	Run/Planned hrs	%	Run/Plan×100	OEE	Improves uptime awareness
283	Performance Calculator	Real-time speed measurement	**PR-LV**	Actual Out/Ideal	%	Act/(Ideal×Hrs)	OEE	Fights hidden cycle losses
284	Quality % Calculator	Good vs total, visible realtime	**PR-LV**	Good/Total	%	Good/Total ×100	FPY	Quality alerts at line

285	Downtime Loss Calculator	Loss visualization → urgency	**PR-LV**	Hrs×Out×Cost	₹	Hrs×Out×Cost	MTTR	Pushes proactive actions
286	Shift Performance Calculator	Shift benchmarking dashboard	**PR-LV**	Plan, Act, Reject	Efficiency%, FPY%	Act/Plan×100	Quality%	Improves accountability
287	KPI Deviation Cost	Impact of KPI fluctuations	**PR-LV**	KPI deviation, Cost/unit	Saved ₹	Dev×Cost×Red%	OEE	Executive interest motivator
288	Plant-wide OEE Gain	Multi-line visibility	**PR-LV**	Curr OEE, Target OEE	Gain ₹/hrs	ΔOEE×Hrs×Value	OEE	For management adoption
289	ROI Calculator	Financial impact of visibility	**PR-LV**	Loss from invisibility, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	All Above	Closing deal tactic
290	FPY Calculator	Measures first-time-right	**PR-QT**	Total, Good	FPY%	Good/Total×100	Quality%	Quality improvement baseline
291	Defect Rate Calculator	Measures % of rejects	**PR-QT**	Defect, Total	Defect%	Def/Total×100	COPQ	Identifies process/drift flaws
292	COPQ Calculator	Cost of rework + rejection	**PR-QT**	Reject×Cost, ReworkCost	₹ Loss	Reject×Cost+Rework	Defect Rate	Biggest cost driver in quality
293	Rejection Reduction Savings	Money saved reducing defects	**PR-QT**	ΔReject%, Out, Cost/unit	Saved ₹	Δ%×Out×Cost	COPQ	Shows business win of quality
294	Quality Deviation Cost	Loss due to quality variation	**PR-QT**	Deviation (units), Cost/unit	Saved ₹	Dev×Cost×Red%	FPY	Justifies defect detection
295	Complaint Recall Risk	Risk of customer returns/recall	**PR-QT**	Events, Cost/event, Red%	Avoided ₹	Cost×Events×Red%	Traceability	C-level risk visibility

296	Root Cause Time Savings	Faster NC resolution	**PR-QT**	ΔTime, Cost/hr	Saved ₹	ΔTime×Cost	MTTR	Helps faster CAPA closure
297	Audit Prep Cost Calculator	Cost reduction in audits	**PR-QT**	Hrs, Cost/hr, Auto%	Savings	Hrs×Cost×Auto%	Trace	Ensures compliance accuracy
298	ROI Calculator	Financial justification for QMS	**PR-QT**	Quality Loss ₹, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	All above	Deal-closing argument
299	Traceability Loss Calculator	Cost of missing genealogy	**PR-TR**	Incidents, Cost, Red%	Savings ₹	Inc×Cost×Red%	Recall	Key root-cause metric
300	Recall Cost Calculator	Risk of product recalls	**PR-TR**	Cost/event, Events, Red%	Avoided ₹	Cost×Events×Red%	COPQ	Highest risk exposure
301	Audit Prep Cost Calculator	Compliance reporting savings	**PR-TR**	Hrs, Cost/hr, Auto%	Saved ₹	Hrs×Cost×Auto%	QMS	Required for audits
302	Root Cause Time Savings	Faster defect trace-back	**PR-TR**	ΔTime, Cost/hr	Saved ₹	ΔTime×Cost	FPY	Prevents customer shutdowns
303	FPY Calculator	Quality impact of genealogy	**PR-TR**	Total, Good	FPY%	Good/Total×100	Defect Rate	Locates origin of rejects
304	ROI Calculator	Business case for traceability	**PR-TR**	Loss, Red%, Cost	ROI%	Loss×Red%×12/Cost	All	Justifies investment to execs
305	Energy Cost Savings Calculator	Cost saved by optimization	**PR-EN**	kWh, Cost, Red%	Saved ₹	kWh×Cost×Red%	CO_2	CFO-driven calculator
306	Utility Load Efficiency	Efficiency gain in utilities	**PR-EN**	Load Δ, Cost	Saved ₹	ΔkWh×Cost	Energy Saver	Baseline for utility efficiency

307	Peak Load Penalty Avoidance	Prevents penalty billing	**PR-EN**	Peak-Allowed, Penalty×Red%	Saved ₹	ΔkVA×Penalty×Red%	Energy Optimizer	Immediate savings case
308	Energy per Unit Cost Calculator	Energy cost per product	**PR-EN**	EngCost, Output	₹/unit	EngCost/Units	CO_2	Helps product pricing & costing
309	CO_2 Calculator	Emission cost mitigation	**PR-EN**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	ESG compliance proof
310	MTTR Calculator	Downtime energy waste reduction	**PR-EN**	Repair hrs, Events	MTTR	Hrs/Event	MTBF	Down machines still consume
311	MTBF Calculator	Stability improves efficiency	**PR-EN**	Run hrs, Events	MTBF	Hrs/Event	MTTR	Avoids energy spike events
312	ROI Calculator	Energy ROI justification	**PR-EN**	Eng Loss ₹, Red%, Cost	ROI%	Loss×Red%×12/Cost	All	CFO decision threshold
313	FPY Calculator	Good assemblies 1st time	**PR-AE**	Total, Good	FPY%	Good/Total×100	Quality%	Direct indicator of assembly accuracy
314	Defect Rate Calculator	Incorrect assembly defects	**PR-AE**	Defect, Total	%	Def/Total×100	COPQ	Defects = rework + lost hrs
315	Assembly Error Cost Calculator	Cost per wrong fitment	**PR-AE**	Errors/mo, Cost/error, Red%	Saved ₹	Err×Cost×Red%	FPY	Business impact of errors
316	Cycle Time Variance	Variation from takt	**PR-AE**	Cycle vs Std(sec)	Δ Time	Cycle–Std	Line Balancing	Slows down whole line
317	Line Balancing Efficiency	Workload imbalance measure	**PR-AE**	Avail vs Actual Cycle	Efficiency%	Ava/Actual×100	Cycle Time	Avoids bottlenecks

318	Root Cause Time Savings	Faster error identification	**PR-AE**	ΔTime, Cost/hr	Saved ₹	ΔTime×Cost	FPY	Quicker Poka-Yoke action
319	Andon Escalation Savings	Faster issue escalation cost impact	**PR-AE**	ΔTime, Cost/min, Events	Saved ₹	ΔTime×Cost×Events	MTTR	Prevents long stoppages
320	ROI Calculator	Financial impact of reducing assembly errors	**PR-AE**	Loss, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	All	CFO approval driver
321	Plant-wide OEE Gain	Scale impact across lines	**PR-SC**	Curr OEE, Target OEE, Value/hr	Gain ₹	ΔOEE×Hrs×Value	OEE	Enterprise-scale benefit
322	Multi-Line Productivity Gain	Expansion of capacity	**PR-SC**	Lines, Thru/line, Imp%, Price	Extra Units + Revenue	Lines×Thru×Imp%	Throughput	Justifies scaling to all plants
323	Centralized Audit Saving	Shared compliance advantage	**PR-SC**	Hours, Cost/hr, Plants	Saved ₹	(Hrs×Cost)×Plants	QMS	Savings multiply across plants
324	Traceability Integration Savings	Multi-site genealogy alignment	**PR-SC**	Incidents, Cost, Red%, Plants	Saved ₹	Inc×Cost×Red%×Plants	Traceability	Multi-facility compliance
325	Data Consolidation Savings	Efficiency of central reporting	**PR-SC**	Reports, Cost/hr, Auto%×Plants	Saved ₹	Hrs×Cost×Auto%×Plants	Visibility	Eliminates reporting cost
326	Standardization Cost Avoidance	Avoid repeated system builds	**PR-SC**	Deployment Cost/site, #sites	Saved ₹	Cost/site×Sites	Platform ROI	Largest cost saving
327	Configuration vs Code Savings	No-code configuration ROI	**PR-SC**	DevCost, ConfigCost, Sites	Saved ₹	(Dev-Config)×Sites	Scale ROI	Faster scalable rollout

328	Platform-Level ROI Calculator	Total business ROI	**PR-SC**	Total Loss, Imp%, System Cost	ROI%	Loss×Imp%×12/Cost	All	Justification for enterprise deal
329	ROI Calculator	Overall investment impact	**PE-CXO**	Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	All	They decide funding
330	Module-Level ROI	Value by module	**PE-CXO**	Loss segmentation	ROI%	ΣLoss×Imp%×12/Cost	All	Evaluates which area to digitize first
331	Plant-wide OEE Gain	Business value of uptime	**PE-CXO**	ΔOEE, Hrs, Value/hr	Gain ₹	ΔOEE×Hrs×Value	OEE	Direct profit uplift metric
332	Multi-Line Productivity Gain	Expansion & capacity gain	**PE-CXO**	Lines, Thru, Imp%, Price	Extra Units/₹	Lines×Thru×Imp%	Throughput	Adds revenue without capex
333	Quality Cost Reduction (COPQ)	Scrap + rework savings	**PE-CXO**	Reject Cost, Rework	Saved ₹	Reject×Cost+Rework	FPY	Quality hits margins hard
334	Energy Cost Savings	Utility profitability	**PE-CXO**	kWh, Cost, Red%	Saved ₹	kWh×Cost×Red%	CO_2	CFO priority metric
335	CO_2 Footprint Calculator	ESG reporting	**PE-CXO**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Governs sustainability goals
336	Scaling Cost Avoidance	Multi-site rollout benefit	**PE-CXO**	Cost/site, Sites	Saved ₹	Cost×Sites	Platform ROI	Biggest value lever
337	OEE Calculator	Master KPI monitor	**PE-PH**	A,P,Q	OEE%	A×P×Q	Availability	Key plant scorecard
338	Availability Calculator	Run vs planned usage	**PE-PH**	Run/Plan hrs	%	Run/Plan×100	OEE	Exposure of idle time
339	Performance Calculator	Speed vs ideal	**PE-PH**	Actual/Ideal	%	Act/(Ideal×hrs)	OEE	Tackles micro-loss

340	Quality % Calculator	Realtime line quality	**PE-PH**	Good/Total	%	Good/Total ×100	FPY	Reduces rework
341	Downtime Loss Calculator	Financial urgency	**PE-PH**	Hrs×Out×Cost	₹	Hrs×Out×Cost	MTTR	Drives maintenance focus
342	Cycle Time Variance	Takt improvement	**PE-PH**	Cycle–Std	Variance	ΔTime	Bottleneck	Improves throughput
343	Shift Performance Calculator	Shift-level benchmarking	**PE-PH**	Plan, Act, Reject	Eff%	Act/Plan×100	Quality%	Accountability of teams
344	ROI Calculator	Plant improvement impact	**PE-PH**	Loss, Imp%, Cost	ROI%	Loss×Imp%×12/Cost	OEE	For business case sign-off
345	MTTR Calculator	Tracks speed of repair	**PE-MT**	Repair hrs, Events	MTTR	Hr/Event	MTBF	Their primary KPI
346	MTBF Calculator	Reliability of assets	**PE-MT**	Run hrs, Failures	MTBF	Run/Event	MTTR	Defines PM strategies
347	Breakdown Cost Calculator	Cost of each fault	**PE-MT**	Same	Lost ₹	Hrs×Out×Cost	Downtime	Shows urgency to fix
348	Downtime Loss Calculator	Daily downtime impact	**PE-MT**	Hrs×Out×Cost	₹	Hrs×Out×Cost	OEE	Prioritizes critical assets
349	Root Cause Time Savings	Time gained from faster RCA	**PE-MT**	ΔTime×Cost/hr	Saved ₹	ΔTime×Cost	FPY	Helps preventive actions
350	Spare Parts Cost Impact	Cost of poor inventory mgmt	**PE-MT**	Carrying, Stock-out Cost	Saved ₹	ΔStock×Carry%	MTTR	Balance cost vs availability
351	Predictive Win ROI	ROI of zero-unplanned downtime	**PE-MT**	Loss from breakdowns, Red%, Cost	ROI%	Loss×Red%×12/Cost	MTBF	Justify predictive analytics

352	Energy Waste in Downtime	Energy loss while idle	**PE-MT**	kWh idle×Cost	Lost ₹	kWh×Cost	Energy	Often ignored hidden cost
353	FPY Calculator	Measure FTR success	**PE-QT**	Total, Good	FPY%	Good/Total ×100	Quality%	Their core KPI
354	Defect Rate Calculator	Scrap % visibility	**PE-QT**	Defect, Total	%	Def/Total×100	COPQ	Faster problem detection
355	COPQ Calculator	Cost of poor quality	**PE-QT**	Reject, Rework Cost	₹ Loss	Reject×Cost+Rework	FPY	CFO justification via Quality
356	Rejection Reduction Savings	₹ saved through quality gains	**PE-QT**	ΔReject%, Output, Cost/unit	Saved ₹	Δ%×Out×Cost	Defect	Links actions to business
357	Quality Deviation Cost	Loss due to drift	**PE-QT**	Dev Units×Cost	Saved ₹	Dev×Cost×Red%	FPY	Supports SPC initiatives
358	Root Cause Time Savings	Faster CAPA closure	**PE-QT**	ΔTime×Cost/hr	Saved ₹	ΔTime×Cost	MTTR	Critical for audit score
359	Complaint Recall Risk	Customer escalation risk	**PE-QT**	Events×Cost	Avoided ₹	Events×Cost×Red%	Traceability	Affects brand trust
360	Audit Prep Cost	Digitized documentation benefit	**PE-QT**	Hrs×Cost/hr×Auto%	Saved ₹	Hrs×Cost×Auto%	QMS	Reduces audit overhead
361	ROI Calculator	Business case via Q-savings	**PE-QT**	Loss, Improve%, Cost	ROI%	Loss×Imp%×12/Cost	All	Helps secure budget
362	Shift Performance Calculator	Shift vs plan tracking	**PE-PR**	Act, Plan	Eff%	Act/Plan×100	Quality%	Daily accountability
363	OEE Calculator	Live production KPI	**PE-PR**	A,P,Q	OEE%	A×P×Q	Availability	Improve shift output

364	Availability Calculator	Runtime vs planned	**PE-PR**	Run/Plan hrs	%	Run/Plan×100	OEE	Reduce idle time
365	Performance Calculator	Speed loss indicator	**PE-PR**	Act/Ideal	%	Act/(Ideal×Hrs)	OEE	Attack micro-stoppages
366	Quality % Calculator	Good parts visibility	**PE-PR**	Good/Total	%	Good/Total ×100	FPY	Immediate defect action
367	Cycle Time Variance	Bottleneck insights	**PE-PR**	ΔTime	Δ	Cycle–Std	Bottleneck	Faster takt improvement
368	Downtime Loss Calculator	Immediate ₹ impact	**PE-PR**	Hrs×Out×Cost	₹	Hrs×Out×Cost	MTTR	Enables prioritization
369	ROI Calculator	Justification for improvement tools	**PE-PR**	Loss, Improve%, Cost	ROI%	Loss×Imp% ×12/Cost	OEE	Proof to management
370	Energy Cost Savings	Cost reduction via efficiency	**PE-ESG**	kWh×₹	Saved ₹	kWh×Cost ×Red%	CO_2	ESG target with financial benefit
371	CO_2 Footprint Calculator	Carbon tracking	**PE-ESG**	kWh, Factor	CO_2 kg	kWh×Factor	Energy	Reporting to regulators
372	Peak Load Penalty Avoidance	Penalty cost reduction	**PE-ESG**	ΔkVA, Penalty	Saved ₹	ΔkVA×Penalty×Red%	Energy	Instant savings path
373	Energy per Unit Cost	Sustainability per product	**PE-ESG**	EngCost, Units	₹/unit	Cost/Units	CO_2	Supports green branding
374	Utility Load Efficiency	Utility behavioral improvement	**PE-ESG**	Load Δ, Cost	Saved ₹	ΔkWh×Cost	Energy	Identifies quick wins
375	Module-Level ROI	ROI via sustainability gains	**PE-ESG**	Energy Loss, Imp%, Cost	ROI%	Loss×Imp% ×12/Cost	Energy	CFO sign-off requirement

376	Centralized ESG Benefit	Multi-plant emissions reduction	**PE-ESG**	CO_2/plant×#Plants	Saved CO_2	CO_2×Plants	Platform Scaling	Real ESG metric for enterprise
377	Configuration vs Code Savings	No-code vs custom-code cost	**PE-IT**	DevCost, ConfigCost, Sites	Saved ₹	(Dev-Config)×Sites	Platform ROI	Faster rollout without dev cycles
378	Standardization Cost Avoidance	Common platform saves rebuilds	**PE-IT**	Cost/site, Sites	Saved ₹	Cost×Sites	Scaling ROI	Multi-plant simplification
379	Centralized Data Consolidation	BI/Data Lake savings	**PE-IT**	Reports×Cost/hr×Auto%	Saved ₹	Hrs×Cost×Auto%	Visibility	Cuts duplication across plants
380	License Optimization Savings	Per-user vs per-site licensing	**PE-IT**	Licenses×ΔCost	Saved ₹	ΔLic×Cost	Infra ROI	Key for budgeting
381	Platform-Level ROI	Total digital transformation impact	**PE-IT**	Loss×Imp%, SysCost	ROI%	Loss×Imp%×12/Cost	All	Board justification metric
382	Integration Cost Avoidance	Avoid custom integration dev	**PE-IT**	APIΔ, DevTime×Cost	Saved ₹	ΔDev×Cost	Platform	Faster time-to-value
383	Deployment Time Reduction	Lower implementation effort	**PE-IT**	ΔWeeks×Cost/week	Saved ₹	WeeksSaved×Cost	Scalability	Enables faster global rollouts
384	Cycle Time Savings	Faster operation pace	**PE-OP**	ΔCycle, Units	Extra Output	Δ×Units	Takt Time	Makes shift performance better
385	Assembly Error Cost	Mistake prevention value	**PE-OP**	Err, Cost/err, Red%	Saved ₹	Err×Cost×Red%	FPY	Fewer rework hours
386	Availability Calculator	Less idle machine time	**PE-OP**	Run/Plan hrs	%	Run/Plan×100	OEE	Direct operator impact

387	Quality % Calculator	Good parts indicator	**PE-OP**	Good/Total	%	Good/Total ×100	FPY	Quick feedback loop
388	Andon Escalation Savings	Faster escalations	**PE-OP**	ΔTime, Cost/min, Events	Saved ₹	ΔTime×Cost×Events	MTTR	Operator empowerment
389	Safety Incident Avoidance	Accidents prevented by alerts	**PE-OP**	Incidents×Cost	Saved ₹	Inc×Cost×Red%	Safety	Critical workforce metric
390	ROI Calculator	Link operator actions to results	**PE-OP**	Loss×Imp%×Cost	ROI%	Loss×Imp%×12/Cost	OEE	Converts productivity into business value
391	Simple OEE Calculator	High-level factory efficiency	**PE-NTI**	A,P,Q	OEE%	A×P×Q	OEE	Quick intro to smart factory
392	Simple FPY Calculator	Good vs bad parts	**PE-NTI**	Good, Total	FPY%	Good/Total ×100	FPY	Easiest quality success metric
393	"Lost Money from Downtime"	Converts downtime to cost	**PE-NTI**	Hrs×Cost/unit×Out/hr	Lost ₹	Hrs×Out×Cost	Downtime	Automatically shows urgency
394	"Cost of Scrap"	Basic defect cost impact	**PE-NTI**	Reject Unit×Cost	Lost ₹	Reject×Cost	COPQ	Simple & relatable damage
395	"Energy Bill Savings"	If your meter reduces X%	**PE-NTI**	kWh×₹×Red%	Saved ₹	kWh×Cost×Red%	Energy	Universal CFO-friendly KPI
396	Business Case Calculator	Mini-ROI explanation	**PE-NTI**	Loss & Cost fields	ROI%	Loss×Imp%×12/Cost	ROI	Educates value step-by-step
397	"Where to Start?" Score	Guided readiness scoring	**PE-NTI**	Simple Yes/No answers	Priority Index	Score model	Visibility	Converts confusion → clarity